

codata
Fundamental Physical Constants

Version 2.5.2

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Preface

Acknowledgment

Chapter 1

Examples

codata Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna

sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

```

program example_in_f
use codata
implicit none
print '(A)', '##### EXAMPLE IN FORTRAN #####'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

#include <stdio.h>
#include "codata.h"
int main(void){
printf("##### EXAMPLE IN C #####\n");
printf("%s\n", "# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n", "# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n", "# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n", "# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}

import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])

```

```
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```


Chapter 2

Man Pages

2.1 `codata(3)`

`codata(3)` Library Calls `codata(3)`

NAME

`codata` - library for fundamental physical constants

LIBRARY

`codata` (`-libcodata`, `-lcodata`)

SYNOPSIS

```
use codata
include "codata.h"
import pycodata
```

DESCRIPTION

`codata` is a Fortran library providing the fundamental physical constants according to CODATA (<https://www.nist.gov/programs-projects/codata-values-fundamental-physical-constants>). A C API allows usage from C, or can be used as a basis for other wrappers. A python wrapper allows easy usage from Python.

The latest `codata` constants (2022) <https://pml.nist.gov/cuu/Constants> were integrated in `stdlib` <https://github.com/fortran-lang/stdlib/releases/tag/> since version 0.7.0. The constants are implemented as derived type which carries the name, the value, the uncertainty and the unit. This library is complementary to the constants defined in the `stdlib` by providing older values for the constants. The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix in their name.

All `codata` (physical) constants are defined as a derived type `codata_constant_type`. All the `codata` constants are provided as double precision reals. The names are quite long and can be aliased with shorter names. The derived type `codata_constant_type` defines 4 members and 2 procedures.

```

type :: codata_constant_type
  character(len=64) :: name
  real(dp) :: value
  real(dp) :: uncertainty
  character(len=32) :: unit
contains
  procedure :: print
  procedure :: to_real_sp
  procedure :: to_real_dp
  generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type

```

A module level interface `to_real` is available for getting the constant value or uncertainty of a constant.

```

o to_real(self, mold, uncertainty) result(r)
  Get the constant value or uncertainty. Warning:
  Some constants cannot be converted to single pre-
  cision sp reals due to the value of the exponents.

o class(codata_constant_type), intent(in) :: self
  Codata constant

o real(sp), intent(in) :: mold
  Should be of type real single or double
  precision

o logical, intent(in), optional :: uncertainty
  Set to true if the uncertainty is required.
  Default to .false..

```

The C API exposes a structure `codata_constant_type` that defines the same members as in Fortran. `to_real` is not exposed in the C API.

```

type, bind(C) :: capi_constant_type
  character(kind=c_char) :: name(65)
  real(c_double) :: value
  real(c_double) :: uncertainty
  character(kind=c_char) :: unit(33)
end type capi_constant_type

typedef struct codata_constant_type{
  char name[65];
  double value;
  double uncertainty;
  char unit[33];
}cct;

```

The Python wrapper encapsulates the members in a dictionary with the keys being `name`, `value`, `uncertainty` and

unit. to_real is not exposed in the Python wrapper.

NOTES

To use codata within your fpm(1) (<https://github.com/fortran-lang/fpm>) project, add the following lines to your file:

```
[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }
```

EXAMPLE

Example in Fortran

```
program example_in_f
  use codata
  implicit none
  print '(A)', '##### EXAMPLE IN FORTRAN #####'
  print '(A)', '# VERSION'
  print *, "version = ", version()
  print '(A)', '# CONSTANTS'
  print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
  print '(A)', '# UNCERTAINTY'
  print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
  print '(A)', '# OLDER VALUES'
  print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
  print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
  print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
  print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program
```

Example in C

```
#include <stdio.h>
#include "codata.h"
int main(void){
  printf("##### EXAMPLE IN C #####\n");
  printf("%s\n", "# VERSION");
  printf("version = %s\n", codata_version());
  printf("%s\n", "# CONSTANTS");
  printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
  printf("%s\n", "# UNCERTAINTY");
  printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
  printf("%s\n", "# OLDER VALUES");
  printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
  printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
  printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
  printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
  return 0;
}
```

Example in Python

```

import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])

```

SEE ALSO

gsl(3), codata(1), fpm(1)

CODATA 2022

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP

- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY

- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90

- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM

- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP

- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT

- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP

- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM

- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
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- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
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- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S

- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE

- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO

- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH

- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT
- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY
- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE

- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV
- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO

- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

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- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018
- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018

- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018
- ATOMIC_UNIT_OF_PERMITTIVITY_2018
- ATOMIC_UNIT_OF_TIME_2018
- ATOMIC_UNIT_OF_VELOCITY_2018
- AVOGADRO_CONSTANT_2018

- BOHR_MAGNETON_2018
- BOHR_MAGNETON_IN_EV_T_2018
- BOHR_MAGNETON_IN_HZ_T_2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- BOHR_MAGNETON_IN_K_T_2018
- BOHR_RADIUS_2018
- BOLTZMANN_CONSTANT_2018
- BOLTZMANN_CONSTANT_IN_EV_K_2018
- BOLTZMANN_CONSTANT_IN_HZ_K_2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018
- CLASSICAL_ELECTRON_RADIUS_2018
- COMPTON_WAVELENGTH_2018
- CONDUCTANCE_QUANTUM_2018
- CONVENTIONAL_VALUE_OF_AMPERE_90_2018
- CONVENTIONAL_VALUE_OF_COULOMB_90_2018
- CONVENTIONAL_VALUE_OF_FARAD_90_2018
- CONVENTIONAL_VALUE_OF_HENRY_90_2018
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_OHM_90_2018
- CONVENTIONAL_VALUE_OF_VOLT_90_2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_WATT_90_2018
- COPPER_X_UNIT_2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2018

- DEUTERON_ELECTRON_MASS_RATIO_2018
- DEUTERON_G_FACTOR_2018
- DEUTERON_MAG_MOM_2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- DEUTERON_MASS_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- DEUTERON_MASS_IN_U_2018
- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018

- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018
- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018
- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018

- ELEMENTARY_CHARGE_2018
- ELEMENTARY_CHARGE_OVER_H_BAR_2018
- FARADAY_CONSTANT_2018
- FERMI_COUPLING_CONSTANT_2018
- FINE_STRUCTURE_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018
- HARTREE_ENERGY_2018
- HARTREE_ENERGY_IN_EV_2018
- HARTREE_HERTZ_RELATIONSHIP_2018
- HARTREE_INVERSE_METER_RELATIONSHIP_2018
- HARTREE_JOULE_RELATIONSHIP_2018
- HARTREE_KELVIN_RELATIONSHIP_2018
- HARTREE_KILOGRAM_RELATIONSHIP_2018
- HELION_ELECTRON_MASS_RATIO_2018
- HELION_G_FACTOR_2018
- HELION_MAG_MOM_2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- HELION_MASS_2018
- HELION_MASS_ENERGY_EQUIVALENT_2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- HELION_MASS_IN_U_2018
- HELION_MOLAR_MASS_2018

- HELION_PROTON_MASS_RATIO_2018
- HELION_RELATIVE_ATOMIC_MASS_2018
- HELION_SHIELDING_SHIFT_2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018
- HERTZ_HARTREE_RELATIONSHIP_2018
- HERTZ_INVERSE_METER_RELATIONSHIP_2018
- HERTZ_JOULE_RELATIONSHIP_2018
- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018

- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018
- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018
- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018
- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018
- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018
- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018

- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018
- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018
- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018
- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018

- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018
- NEUTRON_ELECTRON_MASS_RATIO_2018
- NEUTRON_G_FACTOR_2018
- NEUTRON_GYROMAG_RATIO_2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- NEUTRON_MAG_MOM_2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- NEUTRON_MASS_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_MASS_IN_U_2018
- NEUTRON_MOLAR_MASS_2018
- NEUTRON_MUON_MASS_RATIO_2018
- NEUTRON_PROTON_MAG_MOM_RATIO_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018

- NEUTRON_PROTON_MASS_RATIO_2018
- NEUTRON_RELATIVE_ATOMIC_MASS_2018
- NEUTRON_TAU_MASS_RATIO_2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018
- NUCLEAR_MAGNETON_2018
- NUCLEAR_MAGNETON_IN_EV_T_2018
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- NUCLEAR_MAGNETON_IN_K_T_2018
- NUCLEAR_MAGNETON_IN_MHZ_T_2018
- PLANCK_CONSTANT_2018
- PLANCK_CONSTANT_IN_EV_HZ_2018
- PLANCK_LENGTH_2018
- PLANCK_MASS_2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018
- PLANCK_TEMPERATURE_2018
- PLANCK_TIME_2018
- PROTON_CHARGE_TO_MASS_QUOTIENT_2018
- PROTON_COMPTON_WAVELENGTH_2018
- PROTON_ELECTRON_MASS_RATIO_2018
- PROTON_G_FACTOR_2018
- PROTON_GYROMAG_RATIO_2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- PROTON_MAG_MOM_2018

- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- PROTON_MAG_SHIELDING_CORRECTION_2018
- PROTON_MASS_2018
- PROTON_MASS_ENERGY_EQUIVALENT_2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- PROTON_MASS_IN_U_2018
- PROTON_MOLAR_MASS_2018
- PROTON_MUON_MASS_RATIO_2018
- PROTON_NEUTRON_MAG_MOM_RATIO_2018
- PROTON_NEUTRON_MASS_RATIO_2018
- PROTON_RELATIVE_ATOMIC_MASS_2018
- PROTON_RMS_CHARGE_RADIUS_2018
- PROTON_TAU_MASS_RATIO_2018
- QUANTUM_OF_CIRCULATION_2018
- QUANTUM_OF_CIRCULATION_TIMES_2_2018
- REDUCED_COMPTON_WAVELENGTH_2018
- REDUCED_MUON_COMPTON_WAVELENGTH_2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018
- REDUCED_PLANCK_CONSTANT_2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S_2018
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018
- REDUCED_PROTON_COMPTON_WAVELENGTH_2018
- REDUCED_TAU_COMPTON_WAVELENGTH_2018
- RYDBERG_CONSTANT_2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018

- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2018
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018
- SECOND_RADIATION_CONSTANT_2018
- SHIELDED_HELION_GYROMAG_RATIO_2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
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- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
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- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
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- TAU_NEUTRON_MASS_RATIO_2018
- TAU_PROTON_MASS_RATIO_2018
- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018
- TRITON_PROTON_MASS_RATIO_2018
- TRITON_RELATIVE_ATOMIC_MASS_2018
- TRITON_TO_PROTON_MAG_MOM_RATIO_2018
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- VACUUM_ELECTRIC_PERMITTIVITY_2018
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- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
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- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
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- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014
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- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2014
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- ATOMIC_UNIT_OF_CURRENT_2014
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- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
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- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2014
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- COMPTON_WAVELENGTH_2014
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- CONDUCTANCE_QUANTUM_2014
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- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
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- DEUTERON_G_FACTOR_2014
- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
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- DEUTERON_MASS_IN_U_2014

- DEUTERON_MOLAR_MASS_2014
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MASS_RATIO_2014
- DEUTERON_RMS_CHARGE_RADIUS_2014
- ELECTRIC_CONSTANT_2014
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2014
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2014
- ELECTRON_DEUTERON_MASS_RATIO_2014
- ELECTRON_G_FACTOR_2014
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- ELECTRON_HELION_MASS_RATIO_2014
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- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
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- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
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- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014

- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
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- ELECTRON_VOLT_JOULE_RELATIONSHIP_2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014
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- ELEMENTARY_CHARGE_2014
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- FINE_STRUCTURE_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_2014
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- HARTREE_HERTZ_RELATIONSHIP_2014
- HARTREE_INVERSE_METER_RELATIONSHIP_2014
- HARTREE_JOULE_RELATIONSHIP_2014
- HARTREE_KELVIN_RELATIONSHIP_2014
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- HELION_G_FACTOR_2014
- HELION_MAG_MOM_2014
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- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- HELION_MASS_2014
- HELION_MASS_ENERGY_EQUIVALENT_2014
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
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- HELION_MOLAR_MASS_2014
- HELION_PROTON_MASS_RATIO_2014
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
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- HERTZ_INVERSE_METER_RELATIONSHIP_2014
- HERTZ_JOULE_RELATIONSHIP_2014
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- INVERSE_FINE_STRUCTURE_CONSTANT_2014
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014
- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
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- INVERSE_METER_KILOGRAM_RELATIONSHIP_2014
- INVERSE_OF_CONDUCTANCE_QUANTUM_2014
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- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
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- JOULE_HARTREE_RELATIONSHIP_2014
- JOULE_HERTZ_RELATIONSHIP_2014
- JOULE_INVERSE_METER_RELATIONSHIP_2014
- JOULE_KELVIN_RELATIONSHIP_2014
- JOULE_KILOGRAM_RELATIONSHIP_2014
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014
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- KELVIN_HERTZ_RELATIONSHIP_2014
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- KELVIN_JOULE_RELATIONSHIP_2014
- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014

- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2014
- KILOGRAM_JOULE_RELATIONSHIP_2014
- KILOGRAM_KELVIN_RELATIONSHIP_2014
- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014
- MAG_CONSTANT_2014
- MAG_FLUX_QUANTUM_2014
- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014
- MOLAR_VOLUME_OF_SILICON_2014
- MO_X_UNIT_2014
- MUON_COMPTON_WAVELENGTH_2014
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- MUON_ELECTRON_MASS_RATIO_2014
- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014

- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2014
- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014
- NATURAL_UNIT_OF_MOMUM_2014
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2014
- NATURAL_UNIT_OF_TIME_2014
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- NEUTRON_COMPTON_WAVELENGTH_2014
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2014
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- NEUTRON_G_FACTOR_2014

- NEUTRON_GYROMAG_RATIO_2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014
- NEUTRON_MAG_MOM_2014
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- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
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- NEUTRON_MASS_ENERGY_EQUIVALENT_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_MASS_IN_U_2014
- NEUTRON_MOLAR_MASS_2014
- NEUTRON_MUON_MASS_RATIO_2014
- NEUTRON_PROTON_MAG_MOM_RATIO_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014
- NEUTRON_PROTON_MASS_RATIO_2014
- NEUTRON_TAU_MASS_RATIO_2014
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2014
- NUCLEAR_MAGNETON_2014
- NUCLEAR_MAGNETON_IN_EV_T_2014
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
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- PLANCK_CONSTANT_2014
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- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014
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- PLANCK_MASS_2014
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014
- PLANCK_TEMPERATURE_2014
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- PROTON_CHARGE_TO_MASS_QUOTIENT_2014
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- PROTON_MOLAR_MASS_2014
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- PROTON_NEUTRON_MAG_MOM_RATIO_2014
- PROTON_NEUTRON_MASS_RATIO_2014
- PROTON_RMS_CHARGE_RADIUS_2014
- PROTON_TAU_MASS_RATIO_2014
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- RYDBERG_CONSTANT_2014
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
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- SHIELDED_HELION_GYROMAG_RATIO_2014
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- TAU_MASS_IN_U_2014
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- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014
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- ALPHA_PARTICLE_MASS_IN_U_2010
- ALPHA_PARTICLE_MOLAR_MASS_2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2010
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- BOHR_MAGNETON_IN_EV_T_2010
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- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
- BOLTZMANN_CONSTANT_IN_HZ_K_2010
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010
- CLASSICAL_ELECTRON_RADIUS_2010
- COMPTON_WAVELENGTH_2010
- COMPTON_WAVELENGTH_OVER_2_PI_2010
- CONDUCTANCE_QUANTUM_2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010
- CU_X_UNIT_2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2010
- DEUTERON_ELECTRON_MASS_RATIO_2010
- DEUTERON_G_FACTOR_2010
- DEUTERON_MAG_MOM_2010
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- DEUTERON_MASS_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- DEUTERON_MASS_IN_U_2010
- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010
- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010
- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- ELECTRON_MASS_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ELECTRON_MASS_IN_U_2010

- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010
- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
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- ELECTRON_TRITON_MASS_RATIO_2010
- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
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- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
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- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
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- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
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- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
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- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
- INVERSE_METER_JOULE_RELATIONSHIP_2010
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- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- MUON_MASS_IN_U_2010
- MUON_MOLAR_MASS_2010
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- MUON_PROTON_MASS_RATIO_2010
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- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
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- NEUTRON_MOLAR_MASS_2010
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- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
- PROTON_MUON_MASS_RATIO_2010
- PROTON_NEUTRON_MAG_MOM_RATIO_2010
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- SHIELDED_HELION_MAG_MOM_2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010

- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2010
- TAU_ELECTRON_MASS_RATIO_2010
- TAU_MASS_2010
- TAU_MASS_ENERGY_EQUIVALENT_2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TAU_MASS_IN_U_2010
- TAU_MOLAR_MASS_2010
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- TAU_NEUTRON_MASS_RATIO_2010
- TAU_PROTON_MASS_RATIO_2010
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- TRITON_G_FACTOR_2010
- TRITON_MAG_MOM_2010

- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- TRITON_MASS_2010
- TRITON_MASS_ENERGY_EQUIVALENT_2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TRITON_MASS_IN_U_2010
- TRITON_MOLAR_MASS_2010
- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010
- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
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2.5.2 22 May 2026 codata(3)

2.2 codata(1)

codata(1) User Commands codata(1)

NAME

codata - Command line for codata

SYNOPSIS

codata [OPTIONS] [REGEX_PATTERN ...]

DESCRIPTION

codata is a command line interface which prints all the codata constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

OPTIONS

--year, -y YEAR

Year of the codata constants: 2022, 2018, 2014, 2010

--pattern, -p PATTERN

```
    Regex pattern for filtering the constants.

--value, -a
    Show only the value.

--error, -e
    Show only the uncertainty.

--usage
    Show usage text and exit.

--help Show help text and exit.

--verbose
    Display additional information.

--version
    Show version information and exit.
```

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urban-jost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p 'molar.*gas','electron.*eV'
codata '[B,b]oltzmann.*eV'
```

SEE ALSO

codata(3)

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